

A Multipurpose Public-key Cryptosystem Based Image Watermarking

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Abstract—In order to combine the copyright protection with the content authentication commendably in the digital watermark, this paper proposes an improved error correction coding -based multipurpose method where the authentication watermark and the robust watermark are embedded in the different coefficients of DCT transform domain respectively. In addition, to strengthen the security, the method still utilizes the error correction coding. Through a series of experiment, the invisibility of the method can be proved both from the PSNR value (PSNR=51.2dB) and the histogram equalization. The similarity between the original watermark and extracted watermark is almost above 95%. The proposed method is effective in image authentication and it's robust against some image processing operations, such as low-pass filtering, Gaussian noise, JPEG lossy compression, cropping and histogram equalization.

Key words: multipurpose watermark; scrambling; error correction coding; DCT domain

I. INTRODUCTION

In recent years, Internet system has become popular for transmitting and receiving digital data. But the booms of multimedia industry also create the problems of copyright protection and content authentication. Authorized holders of the multimedia products always want to prevent the illegal distributions and reproductions of the products. On the other hand, receivers of the multimedia products want to assure the integralities and authenticities of the products. So communication security becomes a major concern.

In order to resolve this problem, various kinds of watermarking are proposed, but a few to introduce multipurpose watermarking. Multipurpose watermarking have the function both of the robust watermark and the fragile, which can resist the malicious attacks and locate the area that be altered. It is more effective in some especial instance.

In this paper, a multipurpose watermarking algorithm is proposed, where the authentication watermark and the robust watermark are both embedded in the DCT transform domain. In addition, in order to strengthen the security, the watermarking algorithm still utilizes the RSA public-key cryptosystem and the error correction coding. The advantage of public-key cryptosystem is that no one can destroy the watermark or get the original unwatermarked image without the owner's private key. Through a series of experiment, supportive evidence was provided to demonstrate the proposed method being effective in image authentication and resisting image processing attacks.

II. A New Multipurpose Watermarking Algorithm

A. The Pretreatment of the Watermarks

With the purpose of enhancing the robustness, we process the robust watermark signal as follows.

1). Encoding the robust watermark by Hamming code [5]

We don't expect that the image make mistakes during the process of transmission. To reduce the bit error rate (BER), we can utilize error correction coding which can be calculated as follow:

$$\begin{cases} a_2 = a_6 \oplus a_5 \oplus a_4 \\ a_1 = a_6 \oplus a_5 \oplus a_3 \\ a_0 = a_6 \oplus a_4 \oplus a_3 \end{cases} \quad (1)$$

$$s_1 = a_6 \oplus a_5 \oplus a_4 \oplus a_2 \quad (2)$$

$$s_2 = a_6 \oplus a_5 \oplus a_3 \oplus a_1 \quad (3)$$

$$s_3 = a_6 \oplus a_4 \oplus a_3 \oplus a_0 \quad (4)$$

Where “ $a_6a_5a_4a_3$ ” are information bits, “ $a_2a_1a_0$ ” are supervise bits and “ $S_1S_2S_3$ ” are parity bits. We can use “ $S_1S_2S_3$ ” to correct the wrong bit.

2) Encrypting the encoded watermark using the RSA public-key cryptosystem

Considering the safety of the watermark algorithm, we utilize the public key to encrypt the robust watermark which is encoded before. To calculate easily, we choose the keys: public-key (n, e) = (143, 7), private-key (n, e) = (143, 103). If we calculate it directly, the result in-between is so big that it maybe overflows. So we use the formula (5) to instead of, which can increase the speed of the calculation.

$$(a * b) \bmod n = [(a \bmod n) * (b \bmod n)] \bmod n \quad [4] \quad (5)$$

3) Using scrambling to enhance the error correction rate

Hamming code just can correct one bit mistake, so we can combine it with scrambling [5] to disperse concentrative mistakes into the entire image, which enhance the rate of the error correction. We choose parameter k as a key, and scramble the watermark processed above as

$$\begin{bmatrix} x_{t+1} \\ y_{t+1} \end{bmatrix} = \begin{pmatrix} 1 & 1 \\ k & k+1 \end{pmatrix} \begin{bmatrix} x_t \\ y_t \end{bmatrix} \quad (6)$$

B. The Data Embedding Method

Considering both robustness and Imperceptibility, we divide the original image into 8×8 blocks, and embed the robust watermark into the DCT middle frequency coefficients, then we get the matrix $B_i (i = 1, 2, \dots, 32)$. Modify $B_i(x1, y1)$ and $B_i(x2, y2)$ according to formula (7) and (8) as follow.

$$\begin{cases} \text{If } B_i(x1, y1) < B_i(x2, y2), \\ B_i(x1, y1) = B_i(x1, y1) + \alpha\omega_i \\ B_i(x2, y2) = B_i(x2, y2) - \alpha\omega_i \end{cases} \quad (7)$$

$$\begin{cases} \text{Or else, } B_i(x1, y1) \geq B_i(x2, y2) \\ B_i(x1, y1) = B_i(x1, y1) - \alpha\omega_i \\ B_i(x2, y2) = B_i(x2, y2) + \alpha\omega_i \end{cases} \quad (8)$$

Moreover, for the frangibility of the watermark, the algorithm embeds the fragile watermark into the DCT high frequency coefficient (7, 7). It can be written as:

$$B_i(i, j) = |B_i(i, j)| + \alpha\omega_i \quad (9)$$

And $\alpha = 1$.

Finally, DCT coefficients are converted into spatial domain through the inverse transform, and then the watermarked image is obtained.

C. Watermark Extracting

We divide the watermarked image into 8×8 blocks to get the matrix $B_i (i = 1, 2, \dots, 32)$. And we get two middle frequency coefficients as $B_i(x1, y1)$ and $B_i(x2, y2)$, then extract the watermark according to formula (10).

$$\omega_i = \begin{cases} 1 & \text{if } B_i(x1, y1) > B_i(x2, y2) \\ 0 & \text{if } B_i(x1, y1) \leq B_i(x2, y2) \end{cases} \quad (10)$$

Afterward, we get the robust watermark using the inverse scrambling, the Hamming decoding and the private-key.

Finally, we extract the fragile watermark as follows:

$$\omega_i = \begin{cases} 1 & \text{if } B_i(7, 7) > 1 \\ 0 & \text{if } B_i(7, 7) \leq 1 \end{cases} \quad (11)$$

Watermarked image, robust watermark and fragile watermark are shown as follow.

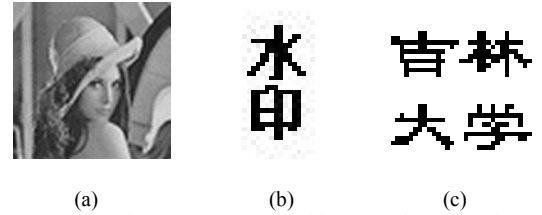


Figure 1. Watermarked image and watermarks
(a) watermarked image; (b) robust watermark;
(c) fragile watermark

III. Simulation and Performance Evaluation

To evaluate the performance of the proposed method, “Lena” with the size of 256×256 is used as the test image. The performance is evaluated according to the imperceptibility, the robustness and verification.

A. Imperceptibility

The image degradation level resulting from noise is evaluated by peak-signal-to-noise-ratio (PSNR) given by

$$PSNR = 10 \log_{10} \left[\frac{255 * 255}{\frac{1}{m * n} \sum_{i=1}^m \sum_{j=1}^n [I(i, j) - I^w(i, j)]^2} \right] \quad (12)$$

Where $I(i, j)$ and $I^w(i, j)$ denote the noise corrupted image and the original image respectively. The PSNR after embedding is 51.2 dB. Moreover, we can see the differences between the original image and the watermarked image more clearly from the histogram, as shown in Figure 2. Come to a conclusion, the visual quality of the watermarked image is maintained.

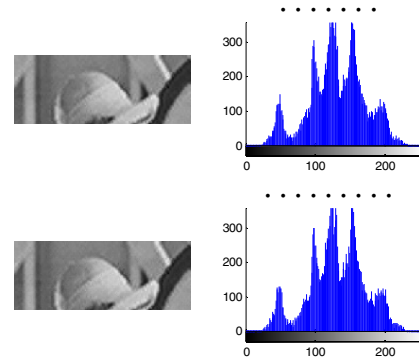


Figure 2. Histograms of the watermarked image and the original image

B. Robustness

We conducted a number of experiments to test the performance of the proposed watermarking algorithm.

Figure 3 shows the watermarked “lean” image with different kinds of attacks and their corresponding best extracted watermarks respectively.

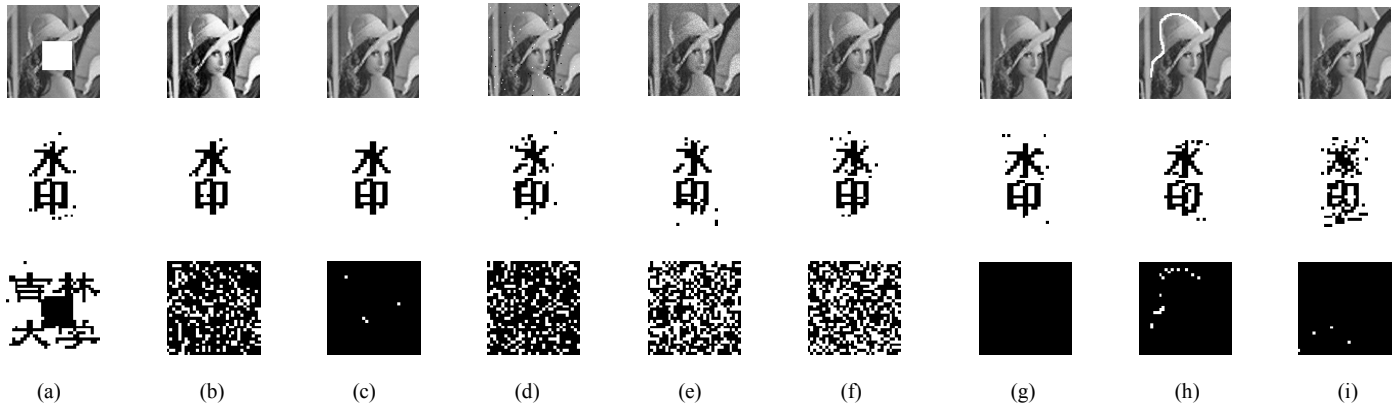


Figure 3. Watermarked image with different attacks .(a) cropping; (b) histogram equalization; (c) low-pass filtering; (d) 1% salt-pepper noise; (e) product noise; (f) Gaussian noise 0.3; (g) Q=50, JPEG; (h) altered; (i) rotation 0.2°

From the Figures shown above, we can see the proposed watermarking algorithm is robust to most of the image processing attacks. Our watermarking can resist any signal processing or data compression as long as the fundamental frequency feature of the watermarked image does not change drastically after processing. Due to the progressive feature of the image transformation, the resulting watermark only suffers different degree of blurring but no serious distortions. Moreover, because the algorithm combines with the error correction coding, the robust watermark is extracted more clearly. Figure 4 shows that the bit error rate is reduced after error correction coding. While in the aspect of authentication, the fragile watermark can not be extracted after signal processing, but it can locate the area which is attacked maliciously, as shown in Figure 3 (a) and Figure 3 (h).

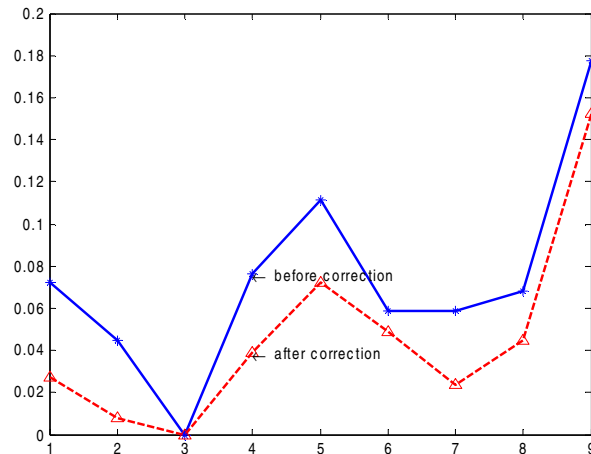


Figure 4. The BER curve (the solid line denote the BER before correction, and the dashed denote the BER after correction)

TABLE I. SIMILARITY BETWEEN ORIGINAL WATERMARK AND DETECTED WATERMARK AFTER VARIOUS ATTACKS

Attack types	Similarity	Attack types	Similarity
cropping	0.9727	Gaussian noise 0.3	0.9629
histogram equalization	0.9922	Q=50,JPEG	0.9766
low-pass filtering	0.9992	altered	0.9551
1% salt-pepper noise	0.9473	rotation0.2°	0.8477

We have tested the robustness against different signal processing operations like Gaussian noise addition 0.3, salt-pepper noise 1%, JPEG compression, cropping etc. The proposed method has shown good performance against many of the attacks as shown in Table 1. The similarity between original watermark and detected watermark is almost above 0.95.

IV. Conclusion

An effective multipurpose image watermarking based on discrete cosine transform domain is proposed in this paper. In the proposed method, the authentication watermark is embedded in the high frequency coefficient and the robust watermark is embedded in the middle frequency coefficient. Moreover, the method also utilizes the RSA public key cryptosystem which can increase the security level because hackers cannot catch our key parameters from the open network. The experiments prove the method proposed is capable to provide a reliable watermark protection even the watermarked image suffered from malicious attacks and locate the altered area efficiently.

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